206-94909E

# Instruction Manual
# S-1700
## THERMOELECTRIC CELL HOLDER

**FOR**
**UV-1600 Series**
**BioSpec-1601　Multispec-1501**
**UV-1700　UV-1800**
**UV-2400/2500 Series**
**UV-3600**

| |
|---|
| Read the instruction manual thoroughly before you use the product. Keep this instruction manual for future reference. |

SHIMADZU CORPORATION
ANALYTICAL & MEASURING INSTRUMENTS DIVISION
KYOTO, JAPAN

# Introduction

## Read this manual thoroughly before using the instrument.

Thank you for purchasing this instrument. This manual describes installation, operation, hardware validation, cautions for use, and details on accessories and options. Read the manual thoroughly before using the instrument. Use the instrument in accordance with the manual's instructions. Keep this manual for future reference.

## IMPORTANT

- If the user or usage location changes, be sure this Instruction Manual is always kept together with the product.
- If this documentation or the warning labels on the instrument become lost or damaged, promptly obtain replacements from your Shimadzu representative.
- To ensure safe operation, read the Safety Instructions before using the instrument.
- To ensure safe operation, contact your Shimadzu representative if product installation, adjustment, or reinstallation (after the product is moved) is required.

# Safety Instructions

To ensure safe operation of the instrument, read these Safety Instructions carefully before use.
Observe all of the WARNINGS and CAUTIONS described in this section. They are extremely important for safety.

In this manual, warnings and cautions are indicated using the following conventions:

| | |
|---|---|
| **WARNING** | Indicates a potentially hazardous situation which, if not avoided, could result in serious injury or possibly death. |
| **CAUTION** | Indicates a potentially hazardous situation which, if not avoided, may result in minor to moderate injury or equipment damage. |
| **NOTE** | Emphasizes additional information that is provided to ensure the proper use of this product. |

## Installation Precautions

To ensure safe operation, contact your Shimadzu representative for installation, adjustment, or re-installation after moving the instrument to a different site.

## In an Emergency (Or During a Power Outage)

In an emergency (or during a power outage), turn OFF the power switch on the back side of the controller.
Turn OFF the power switch of the spectrophotometer main unit to which the product is connected, also.

# Warning Labels on Equipment

For safety operation, warning labels are affixed where special attention is required.
Should any of these labels peel off or be damaged, obtain replacements from Shimadzu Corporation.

WARNING

HOT SURFACE

The temperature of cell lid becomes almost same as the set temperature during the measurement. There is a risk of burn if touching the cell lid while it is hot. When using the cell at high temperature measurement, check that the cell has been cooled down before mounting or removing the cell.

In addition to it, during measurement, do not remove cell by pulling lever.

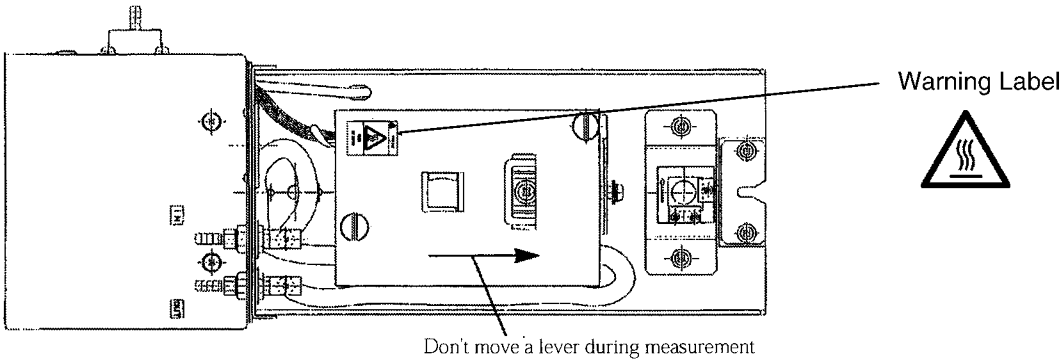


Fig1 Warning Label No.1

WARNING

LISK OF ELECTRIC SHOCK

Risk of electric shock.

Before replacing the fuse, turn off the power and unplug the power cable.

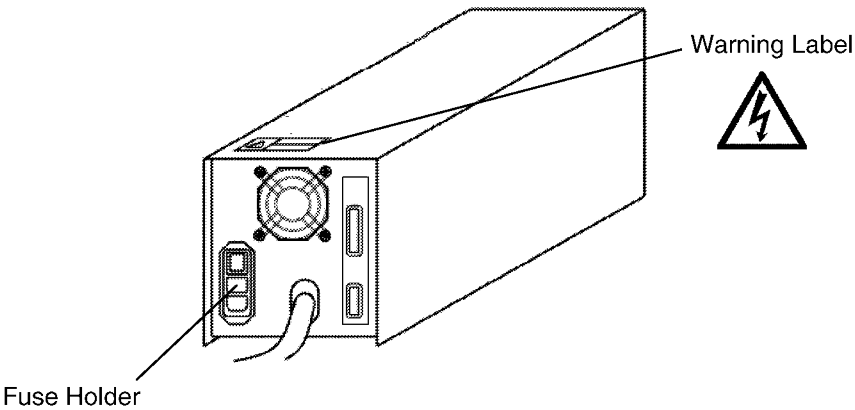


Fig2 Warning Label No.2

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# Symbols Indicated on the unit

| Symbol | Contents |
|---|---|
| ~ | Indicates current (a.c.) |
| \| | Indicates Power ON |
| ○ | Indicates Power OFF |

# Regulatory Information

For Europe:
The product complies with the requirements of the EMC Directive 89/336/EEC amended by 92/31/EEC, 93/68/EEC, and Low Voltage Directive 73/23/EEC amended by 93/68/EEC.

Product name : Thermoelectric Cell Holder

Model Name : S-1700

Manufacture : SHIMADZU CORPORATION
ANALYTICAL INSTRUMENTS DIVISION

Address : 1, NISHINOKYO-KUWABARACHO,
NAKAGYO-KU, KYOTO, 604-8511, JAPAN

Authorized Representative In EU : SHIMADZU Deutschland GmbH

Address : Albert-Hahn-Strasse 6-10, D-47269
Duisuburg F.R.Germany

# Electro Magnetic Compatibility

**Note** Descriptions of this section are only applied to the model for EU (European union) market : 206-56000-38

This instrument complies with European standard EN61326-1 which is amended at 1998

Emission : Industrial Location

Immunity : Industrial Location

1) Electro Magnetic Emission

This instrument can use in industrial environments.

The test specifications are stated below.

1. Radiated Emission ( EN61326-1 : 1997 + A.1998 Class B )
2. Conducted Emission ( EN61326-1 : 1997 + A..1:1998 Class B )

**Note** When a electro magnetic disturbance occur to those instrument being used close to this prodest. Take appropriate distance between instruments to eliminate to eliminate the disturbance.

2) Immunity to Electro Magnetic Interference

This instrument can use in industrial environments.

The test specifications are started below.

1. Radiated Susceptibility ( EN61000-4-3 : 1996 )
   Field strength : 10V/m, Frequency range : 80-1000MHz, Modulation : 1KHz AM80%

**Note** In this condition, the datum of this instrument may be noisy at same frequencies, but the instrument recovered automatically when the electromagnetic radiation has stopped.

2. Power Magnetic Fields ( EN61000-4-8 : 1993 )
   Magnetic Field strength : 30A/m
3. Electrostatic Discharge ( EN61000-4-2 : 1995 )
   Charge Voltage : ±4KV ( Contact ), ±8KV ( Air Discharge )
4. Fast Transients on AC Power Lines ( EN61000-4-4 : 1995 )
   Test Voltage : ±2KV

5. Surge Immunity of AC Power Lines ( EN61000-4-5 : 1995 )
   Test Voltage : ±1KV line to line, ±2KV line to earth
6. Conducted Radio Frequency Immunity of AC Power Lines ( EN61000-4-6 : 1996 )
   Applied Voltage : 3V, Frequency Range : 150k-80MHz, Modulation : 1KHz AM80%
7. Immunity Against Voltage Dip and Interruptions on AC Power Lines ( EN61000-4-11 )

**Note** Compliance to the standard does not ensure that the instrument can work with any level of Electro Magnetic interference stronger that the level tested.
Interference greater than the value apecified in the condition above may cause malfunction of the instrument.
To avoid electro magnetic disturbance, follwing notices are recommended to be followed.

1) Brfore touching the instrument, discharge the electro statics charged in operator's body to ground by touching metallic structure connected to ground.
2) Do not install this instrument in such environment where strong electro magnetic fields are generated near by.

# Product Warranty

Our company provides a warranty on this product, as stated below.

**Details**

**1. Warranty Period:** Please consult your Shimadzu representative for information about the extent of the warranty.

**2. Warranty Description:** If failure occurs for reasons attributable to our company during the warranty period, our company will provide repairs or the replacement of parts without charge (including USB dongles). However, we may not be able to provide identical products in the case of products such as PCs, and their peripherals and parts, which have a short lifespan in the market.

**3. Warranty Exceptions:** The failures caused by the following events are excluded from the warranty, even if they occur during the warranty period.

1) The product is handled in an improper way.
2) Repairs or modifications are performed by companies or people other than our company and our designated companies.
3) This product was used in combination with hardware or software other than those designated by our company.
4) Device failures and damage to data and software, including the basic software, that are caused by computer viruses.
5) Device failures and damage to data and software, including the basic software, that are caused by power failures, including power outages and sudden drops of voltage.
6) Device failures and damage to data and software, including the basic software, that are caused by powering off the device without the proper shutdown procedure.
7) Failures caused by reasons other than the device itself.
8) Failures caused by use in harsh environments, such as in high temperature or humidity, corrosive gas, or vibration.
9) Failures caused by fires and earthquakes or any other act of providence, contamination by radio active substances and hazardous substances, or any other force majeure event including wars, riots, and crimes.
10) Problems occur because the device is transferred or transported after installation.
11) Expendable items and parts
    Note: Recording media such as floppy disks and CD-ROMs are considered expendables.

* If there is a document such as a warranty attached to the product, or there is a separate contract agreed upon that includes warranty conditions, the rules stated in those documents shall be followed.
Warranty periods for products with special specifications and systems are provided separately.

# After-Sales Service and Replacement Parts Availability

**After-Sales Service**

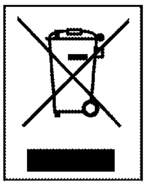
If any problem occurs with this instrument, inspect it and take appropriate corrective action as described in the Section "10.1 Troubleshooting".
If the problem persists, or symptoms not covered in the Troubleshooting section occur, contact your Shimadzu representative.

**Replacement Parts Availability**

Replacement parts for this instrument will be available for a period of seven (7) years after the discontinuation of the product. Thereafter, such parts may cease to available. Note, however, that the availability of parts not manufactured by Shimadzu shall be determined by the relevant manufacturers.

# Action for Environment (WEEE)

## To all user of Shimadzu equipment in the European Union:



WEEE Mark

Equipment marked with this symbol indicates that it was sold on or after 13th August 2005, which means it should not be disposed of with general household waste. Note that our equipment is for industrial/professional use only.

**Contact Shimadzu service representative when the equipment has reached the end of its life. They will advise you regarding the equipment take-back.**

With your co-operation we are aiming to reduce contamination from waste electronic and electrical equipment and preserve natural resource through re-use and recycling.
Do not hesitate to ask Shimadzu service representative, if you require further information.

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# Contents

6 How to Use Optional PC Software (Temperature Control in PC)



Section 7 Maintenance and Inspection

# Section 1 Before Operation (Installation)

Contents

| 1.1 | Parts Inspection |
|---|---|

This system consists of the parts listed below. After unpacking, check that the parts are included properly.

Table.1.1 Standard Accessories

| | Parts Name | Part No. | Q'ty | Remarks |
|---|---|---|---|---|
| 1 | Temperature controller | 206-55040-93<br>or<br>206-55040-94 | 1<br><br>1 | 100～120V(-91)<br><br>220～240V(-38) |
| 2 | Sample compartment | 206-56005-93<br>or<br>206-56005-94 | 1 | 100～120V(-91)<br><br>220～240V(-38) |
| 3 | Signal cable | 206-56097-91 | 1 | 25 pin |
| 4 | Stirrer controller | 208-97250-21<br>or<br>208-97250-24 | 1<br><br>1 | 100～120V (-91)<br><br>220～240V (-38) |
| 5 | AC power cable | 071-60814-01<br>or<br>071-60814-05 | 1<br><br>1 | 100～120V (-91)<br><br>220～240V (-38) |
| 6 | Power fuse | 072-02004-19<br>or<br>072-02004-17 | 2<br><br>2 | 2A type T (-91)<br><br>1.25A type T(-38) |
| 7 | Stirrer tip | 046-00619-01 | 2 | 4×6mm |
| 8 | Cell Cover Assy | 206-56224-91 | 1 | |
| 9 | Cell Spacer | 206-56225 | 1 | |
| 10 | Silicon Tube $\phi$8×4 | 016-31350-74 | 1 | 4m |
| 11 | Vinyl Tube $\phi$13×10 | 016-31310 | 1 | 4m |
| 12 | Adapter | 035-61574-01 | 3 | |
| 13 | Hose Band ※1) | 072-60320-01 | 6 | |
| 14 | Hose Clamp ※2) | 037-61018 | 6 | |
| 15 | Mask | 206-55995-02 | 1 | Use this at using<br>115B-QS-10<br>Refer to chapter 5.3 |
| 16 | Instruction Manual | 206-94909 | 1 | English |
| 17 | Instruction Manual | 206-94908 | 1 | Japanese |

※1) use to band the hose at the front panel of sample compartment.

※2) use to clamp the hose at the connector of water circulation unit.

## 1.2 Installation Site

Install this cell holder in a site satisfying the following conditions so that it can be used stably with its full performance for a long time.

Even within the guarantee period, our company is not responsible for any deterioration in the performance and any damage in the structure caused by use in a site not satisfying the following conditions.

- The room temperature during operation shall be 15 to 35℃.
- The humidity during operation shall be 45 to 80%.
- The cell holder shall not be exposed to direct sunlight.
- Not only strong vibration but also continuous minute vibration shall not be generated.
- Strong magnetic fields and electromagnetic waves shall not be present.
- Erosive gas as well as organic and inorganic gas having the absorptivity in the ultraviolet area shall not be present.
- Much dusts shall not be present.

The size of this instrument (temperature control unit) is 140mm(W)×340mm(D)×170mm (H). Necessary minimum installation floor area for temperature controller is 200mm(W)×400mm (D) in addition to the minimum area for UV-VIS Spectrophotometer. Since there is a fan for ventilation at the rear of the temperature controller unit, leave enough space and don't put anything between the wall and fan.

The weight of this instrument is 11 kg. Select an installation flat surface capable of supporting the weight of this instrument and UV-VIS Spectrophotometer.

### 1.2.1 Ventilation

When using organic solvents as measurement samples, perform sufficient ventilation in the installation site.

### 1.2.2 Cleaning the Instrument

Use the temperature controller cover and the sample compartment in the clean status.

When cleaning them, wipe them lightly with soft cloth soaked in a little water or diluted detergent.

Do not wipe them with too wet cloth. Do not pour directly any detergent to them.

**CAUTION** Do not spill water or organic solvent on the cell holder.
Such spill may cause electrical failures or mechanical failures.

## 1.3 Before Starting Measurement

### 1.3.1 Power Requirement

The power consumption of this cell holder is 110 VA. Prepare a power supply whose capacity is 110 VA or more. The allowable voltage fluctuation range is within ±10%. If the voltage fluctuation of the power supply to be used exceeds this range, use a voltage stabilizer.

### 1.3.2 Grounding

The power cable of this cell holder is the three-wire type including a grounding wire. Insert the power plug into the three-wire type power outlet, and securely perform grounding.

| NOTE | If the power supply outlet of 2-line type is used, prepareyhe grounding adapter of 3-line type, be sure to ground from grounding terminal of it. |
|---|---|

### 1.3.3 Connecting the Cable

(1) Confirm that the power switch of the temperature controller is OFF (in which "O" is pushed in).
(2) Confirm the supply voltage indicated on the rear of the main unit.
(3) Connect the power cable offered as an accessory to the connector provided on the rear of the temperature controller.
(4) Connect the power cable to the power outlet.
(5) Attach the sample compartment to the spectrophotometer main unit, then fix it by tightening knurled screws provided under the sample compartment. (refer to Fig.1.1)
(6) Connect the round connector cable from the temperature controller to the round connector provided on the side of the sample compartment. (refer to Fig.1.2)
(7) Connect the controller and the sample compartment with s signal cable (25-pin connector cable) offered as an accessory. (refer to Fig.1.2)
(8) Connect the stirrer connector from the front of the sample compartment to the stirrer controller.
(9) Set to ON the power switch of the temperature controller.
(10)Connect the power cable of the stirrer controller.*

*There is no power switch on stirrer controller. If you connect power cord, input power supply and be able to control rotation speed. In case of useless, disconnect the power cable to the power supply outlet.

### 1.3.4 Connecting the Cooling Water Plumbing (Constant-temperature Circulation Unit)

This unit performs the temperature control by peltier element. So, when controlling the temperature with this instrument, it is necessary to circulate the cooling water to cool the Peltier element. The specifications for water circulation are shown below.

| | | |
|---|---|---|
| Water temperature for circulation | : | 20±2℃ |
| Flow rate | : | 4.8 L/min or greater |
| Hose connector O.D. | : | Inner Diameter $\phi$ 8, 10, 12 mm |

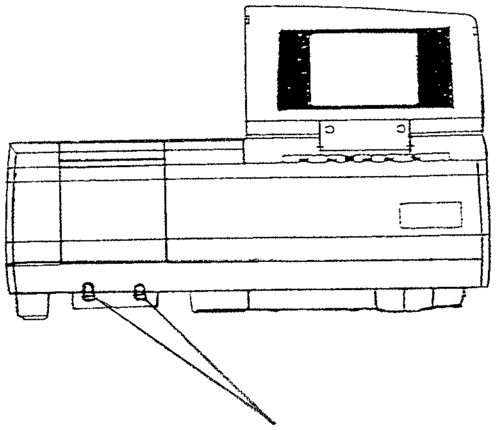



Fig.1.1 Fix the Sample Compartment

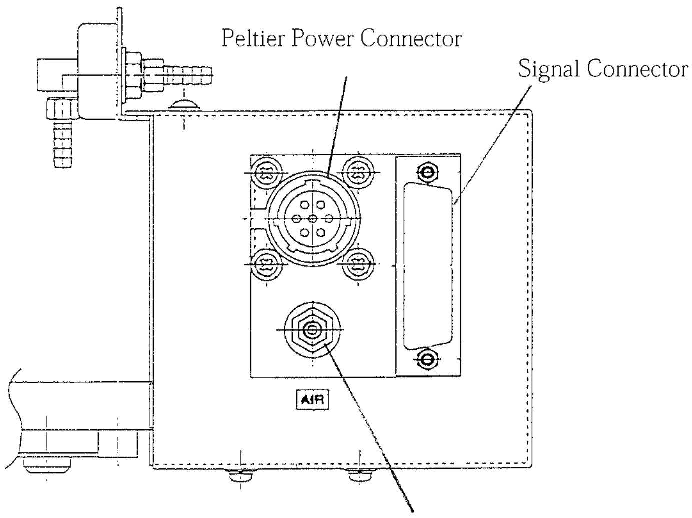



Fig.1.2 Method of Connecting Cables

### 1.3.5 Connecting $N_2$ Purge Gas Plumbing

In case that under 10℃ measurement points are included, it is necessary to connect $N_2$ purge gas (or dry air) for preventing a dew drop of cell sureface.

Gas Quantity ： about 3L／minute (Max 5L / minute)

| | |
|---|---|
| CAUTION | About purge quantity…<br>Not exceed the Limit quantity (5L/min). In case of connecting directly from room plumbing. There is possibility that the temperature of purge gas is higher than room temperature and can not control successfully of especially low temperature. Set correctly the connection and quantity of purge gas. |

| | |
|---|---|
| CAUTION | If dew drop should stick….<br>Even if purge setting is done correctly according to above explanation, there is possibility to stick a dew drop to cell surface. Check following items.<br>① In case that room temperature is over 30℃, use $N_2$ gas-cylinder.<br>(not use room plumbing)<br>② Does the purge tube bend on the way?<br>③ Measure with more slow Ramp Rate. |

### 1.3.6 Turning on the Power and Initialization

When the temperature controller is powered on, the temperature is displayed on the sheet key of the temperature controller in order of [Set Temperature]→ [Current Temperature]→ [Set Temperature](repetition). If not displayed, the cables may not have been connected properly. Check the cable connection.

# Section 2 Description of Each Part

Contents

# 2.1 Appearance

## 2.1.1 Temperature Controller Unit

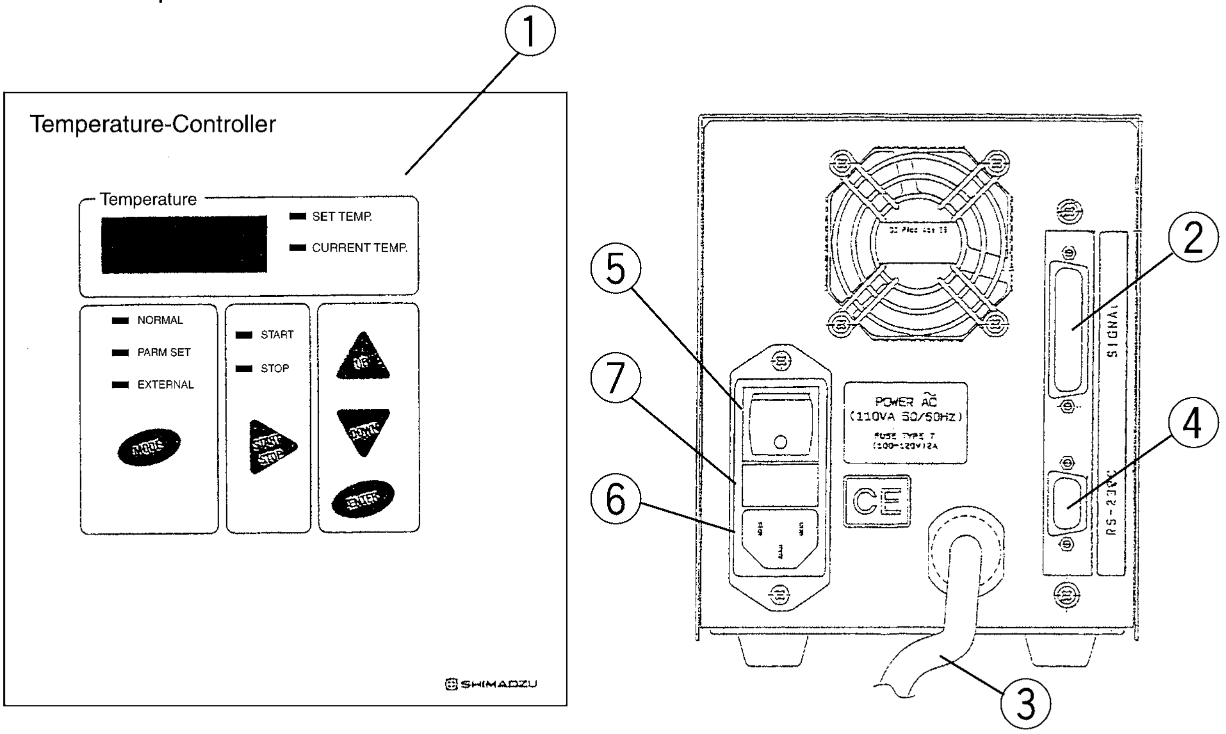



Fig.2.1 Front and Rear of Temperature Controller

① Sheet key panel

Switches the mode, sets the temperature program and Starts/Stops temperature control manually.

4 places-digid LED is provided to display the temperature.

② Signal connector

Inputs and outputs the temperature control signal.

③ Peltier power cable

Connected to the sample compartment, and supplies the power for temperature control to the sample compartment.

④ PC communication connector

Connected to a personal computer, and enables control of the cell holder by Tm Analysis software (Option). (This connector is not used in the stand-alone type.)

⑤ Power switch

Turns on (I) and off (O) the power.

⑥ AC power connector

Connected to the power cable offered as an accessory, and supplies the power from the AC power outlet.

⑦ Fuse holder

Use 2A fuses in the 100 to 120V area. Use 1.25A fuses in the 220 to 240V area.

### 2.1.2 Sample Compartment

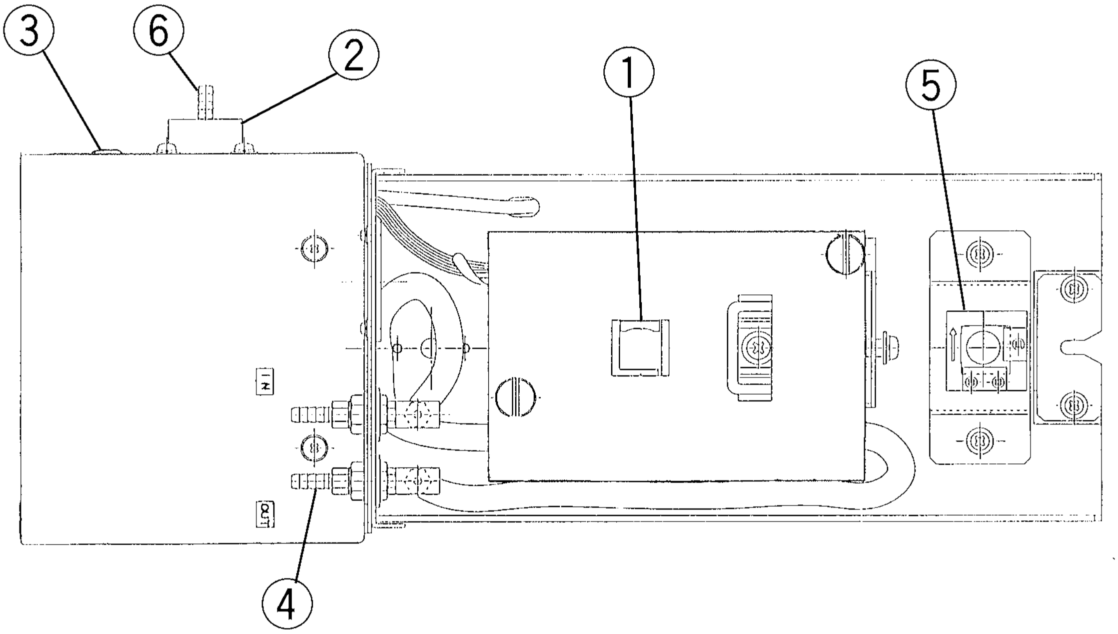



Fig.2.2 Sample Compartment

① Cell holder

Mounts a commercial square cell of 10×10 mm. (about cell type, refer to Chapter 3)

② Peltier power connector

Supplies the power for temperature control from the temperature controller.

③ Signal connector

Connected to the temperature controller, and receives and sends the temperature control signal.

④ Hose connector

Used to circulate the cooling water.

⑤ Reference cell holder

Used to install a sample on the reference side. (Temperature control and stirrer control is not available.)

⑥ Purge connector

Used to connect the purge gas tube ($N_2$ gas or Dry air) to avoid being stained a dew drop to surface of cell.

| | |
|---|---|
| NOTE | In this system, cell is inserted to cell holder with attaching the private cell cover (No ⑧ in table 1.1) for preventing an evaporation of sample quantity.<br>So, it is possible to use only recommendation cells. (refer to Chapter 3) |

### 2.1.3 Stirrer Unit

When using the stirrer, connect the connector from the front panel of the sample compartment to the dedicated stirrer controller offered as an accessory by using the following procedure.

1) Connect the cable from the front panel of the sample compartment to the connection port in the stirrer controller offered as an accessory.
2) Turn on the power of the stirrer controller.
   The input voltage follows to indicated value of stirrer controller.
   ( 100V area : input voltage range is 100～120V)
   ( 200V area : input voltage range is 220～240V)
3) Turn the rotation setting trimmer in the stirrer controller to rotate the stirrer chip.

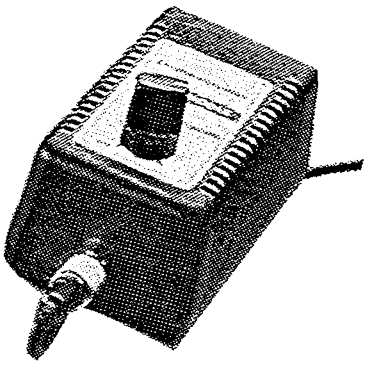



Fig.2.3 Connecting the Stirrer

| 2.2 | Sheet Key of Temperature Controller |
|---|---|

In this cell holder, perform temperature control and measurement by manipulating the sheet key in the temperature controller.

The detailed operating procedures are described in Chapter 4.

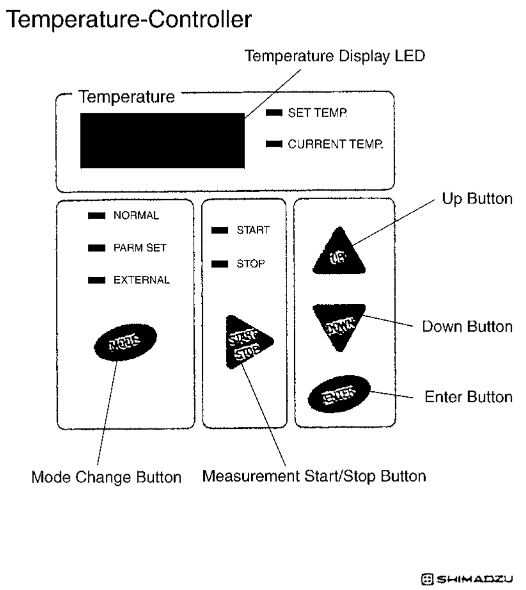



Fig.2.4 Sheet Key

"Mode Display LED"

Displays the current mode. When the mode selector button is pressed, the mode is switched in turn as follows:

"Normal" ———— Performs "Set Mode". This Mode keeps the cell at the temperature set to temperature controller.

"Parm Set" ———— Allows to input measurement parameters to be executed in the temperature controller.

"External" ———— Performs measurement by using Tm Analysis software (option). This mode is not available in this system. When the power of the cell holder is turned on, this mode is automatically selected usually. Press the mode selector button, and select another mode.

"Measurement Start/Stop LED"

Displays the current measurement status.

"Temperature Display LED"

Displays the target temperature*, the current temperature in this order in the 3-second cycle.

"Temperature Display 7-Segment LED"

Displays the temperature in four digits in a range from -10 to +120℃.

"Mode Change Button"

Switches the mode every time it is pressed.

"Measurement Start/Stop Button"

Starts or stops measurement.

"UP Button"

Increases the coefficient of the set temperature, etc. in the measurement parameter setting mode.
This button is available also even while the temperature is in control.

"DOWN Button"

Decreases the coefficient of the set temperature, etc. in the measurement parameter setting mode.
This button is available also even while the temperature is in control.

"ENTER Button"

Determines an input coefficient in the measurement parameter setting mode.

# Section 3 Specifications

# Hardware specifications

<Temperature specifications>

| | |
|---|---|
| Temperature input range | 0～110℃ |
| Temperature display range | -10～120℃ |
| Temperature display unit | 0.1℃ |
| Temperature rate | ±0.1, 0.2, 0.5, 1.0, 2.0, 5.0℃/minute. 12 steps |
| Cell inside temperature accuracy* | Within ±0.25℃ (0～25℃)<br>Within ±1%℃ to Set value (25～75℃)<br>Within ±2%℃ to Set value (75～110℃)<br>case of using at 25℃ room temperature |
| Dependence on room temperature<br>For cell inside temperature accuracy | ±0.01×(t-25)℃<br>t：room temperature |
| Cell inside temperature stability | Within ±0.15℃ |

* Cell inside temperature accuracy is within ±5%℃ to Set value for above 100℃ and within ±0.5℃ about under 0℃.

<Sample compartment>

| | |
|---|---|
| Minimum required sample amount | 400μL (case of using 115B-QS-10*)<br>3.5mL (case of using 110-QS-10*) |
| Cell optical path length | 10 mm |
| Cell material | Quartz |
| Heating/cooling method | Electronic cooling/heating method by<br>Peltier elements |
| Peltier protecting method | Water-cooling method with circulating water |
| Used sensor | Platinum resistance thermometer bulb |
| Purge function | Yes |
| Reference side | Standard cell holder<br>(without temperature control and stirrer function) |
| Stirrer mechanism | Provided (100～1000 rpm) |
| Applicable model | UV-1601 / 1601PC / 1650PC, BioSpec-1601<br>UV-1700, Multi Spec-1501<br>UV-2401PC / 2450, UV2501PC / 2550<br>UV-3600 |

* manufactured by HELLMA

<Temperature controller>

| | |
|---|---|
| Temperature display function | 7 segment LED display |
| Temperature display unit | 0.1℃ |
| Control method | method |
| Temperature setting function | Yes |
| Temperature program function | Yes<br>(Memorizes only one program) |

<Stirrer>

| | |
|---|---|
| Stirring point | One position in center |
| Stirring capacity | Cuvette cell: 0.1～5 mL |
| Number of rotations | 100～1,000 rpm |
| Rotation stability | ±3% |
| Stirring force | 0.15W |
| Size | Control unit: W55×D90×H40 mm |
| Weight | Control unit: Approx. 400g |
| Environmental condition | Control unit: 0～40℃ (Humidity 80% or less) |
| Working voltage | 12 VDC |
| Input power supply | 100 VAC 50/60 Hz (−91)<br>or<br>220～240 VAC 50/60 Hz (−38) |
| Cable length | Power code 2M/connection code 1M |
| Stirrer chip size | 4×6mm |

NOTE: Take care that the stirring capacity varies depending on the cell shape and the sample viscosity.

NOTE: When any stirrer chip other than the specified one is used, the chip may not rotate.

<Usable Cell List>

| Cat No. | Path Length | Required minimum sample vol. |
|---|---|---|
| 110-QS-10 | 10mm | 3.5mL |
| 115B-QS-10* | 10mm | 400μL |

Each cell is manufacted by HELLMA.

＊It is necessary to use together with cell spacer (No ⑨ in table 1.1).

But in this case, it is impossible to use stirrer rotation.

# Section 4 Operation of Temperature Controller

This section describes how to manipulate the sheet key in the temperature controller.

The temperature controller controls the temperature in two modes, the jump mode which maintains a constant temperature and the program mode which controls the temperature in accordance with the input temperature speed, etc. as described below.

Contents

## 4.1 Temperature Jump Mode

In this mode, the cell holder temperature is changed to the set value.
Once the cell holder reaches the set temperature, it is maintained at that temperature.

"Temperature jump setting method"

1) The temperature jump mode is effective when the mode display LED indicates "Normal" on the sheet key screen. If any other mode is selected, press the MODE button on the sheet key to display "Normal".
2) Press the UP button while checking the value displayed on the temperature display LED to display the temperature in the jump destination.
3) Press the START/STOP button to start control toward the set temperature.

NOTE Every time the "UP" button is pressed once, the displayed temperature increases by 0.1℃. When it is pressed and held for several seconds, the temperature increases by 0.3℃. When it is pressed and held further more, the temperature increases by 0.5℃.

NOTE While the temperature display LED indicates "SET.TEMP", the set temperature is displayed.
Even the "start" LED is indicated, it is possible to change the set temperature with "UP / Down" button. However, in case of changing more than ±5℃, once press "Stop" button and after stopping temperature control, change the set temperature.

NOTE In this mode, cell temperature is controlled with ±5℃/min ramp rate. It is impossible to change ramp rate.

| 4.2 | Temperature Program Mode |
|---|---|

In this cell holder, the temperature controller can create and execute a program which controls the temperature in accordance with the temperature increase/decrease speed of cell holder, etc. input by the operator. However, measurement of the absorbance in interlocking with the spectrophotometer is not available.

A created program is stored in the internal memory. When the power is turned on at the next time, the same program can be executed without inputting it again.

"Temperature program input/execution procedure"

1) If measurement is being performed, finish it.
2) Press the "MODE" button until the mode display LED indicates "PARAM.SET".
3) Press and hold the "ENTER" button for several seconds until the temperature display LED becomes ready for receiving input of the start temperature. Input each set value in the following sequence. Every time the "ENTER" button is pressed, each parameter set value input screen appears. Input all parameters.

| | |
|---|---|
| ① Start temperature | : point of starting temperature control |
| ② Start temperature hold time | : times until starting control |
| ③ Temperature speed | : control speed |
| ④ Next temperature | : next control temperature |
| ⑤ Pre-control wait time | : wait time for every control point |
| ⑥ Repeat Times | : repeat times between ① and ④ temperature |

4) Press the START/STOP button to start temperature control.

**Temperature program parameters**

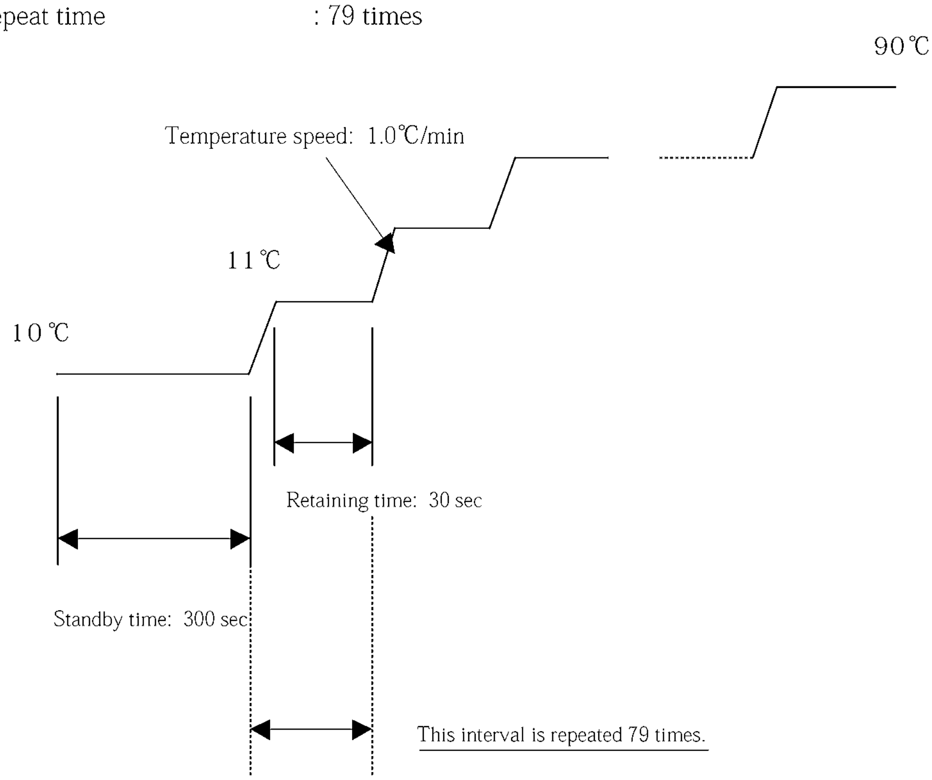
| | |
|---|---|
| ① Start temperature | : Set in the increment of 0.1℃ in a range from 0 to 110℃. |
| ② Start temperature hold time | : Set in the increment of 1 second in a range from 0 to 9999℃. |
| ③ Temperature speed | : Set to either of twelve steps, ±0.1, ±0.2, ±0.5, ±1.0 ±2.0 and ±5.0℃/min. |
| ④ Next temperature | : Set in the increment of 0.1℃ in a range from 0 to 110℃. |
| ⑤ Pre-control wait time | : Set in the increment of 1 sec in a range from 0 to 9,999 sec.<br>When it is set to [-](under [0]), the retaining time is infinite. |
| ⑥ Repeat times | : Set in a range from 0 to 9,999 times.<br>If no.⑤ is set to [-], automatically set to[0]. |

<Example of temperature program creation 1>

- When the temperature is to be controlled in the increment of 1℃ in a range from 10 to 90℃at the temperature speed of 1℃/min with the retaining time of 30 seconds at each interval.

  Set the parameters as follows:

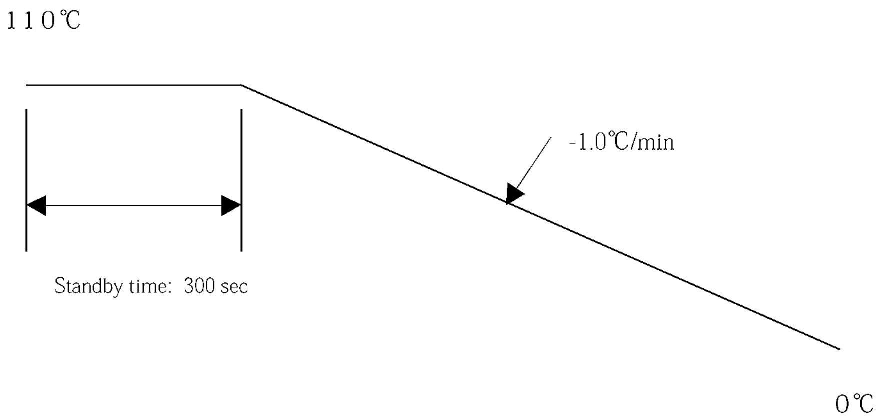
| | |
|---|---|
| Start temperature | : 10℃ |
| Start temperature hold time | : 300 sec |
| Temperature speed | : +1.0℃/min |
| Next temperature | : 11.0℃ |
| Pre-control wait time | : 30 sec |
| Repeat time | : 79 times |



<Example of temperature program creation 2>

- When the temperature is to be decreased at the temperature speed of -1℃/min in a range from 110 to 0℃

| | |
|---|---|
| Start temperature | : 110.0℃ |
| Start temperature hold time | : 300sec |
| Temperature speed | : -1.0℃/min |
| Next temperature | : 109.0℃ |
| Pre-control wait time | : 0 sec |
| Repeatability | : 109 times |



## 4.3 Error Message in Temperature Control

If anything troubles among this system happen, the error code indicated unusual displays in 7-segment LED.

Please confirm items as follows, check that the system works correctly. If the system should not be corrected, contact with service department of Shimadzu group.

| Error Message | Cause | Confirmation item |
|---|---|---|
| E1 | Circulation water for protecting peltier is missing. | Confirm that a cooling water of sample-conpartment front panel should circulate. Confirm that the tube for cooling water should not bend on it way. Confirm that water temperature should be within 20±2℃. |
| E3 | Unusual Heating of peltier unit happens. | Once turn off the power of controller and turn on the power again. If the error happens repeatedly, contact with service department of Shimadzu group. |
| E132 | System can not find the temperature sensor. | Confirm whether the signal cable of side panel should connect correctly. Fix the cable firmly with screw of cable contact. |

# Section 5 Handling of Cell

This section describes how to attach a cell to the cell holder and how to clean a cell.

Contents

| 5.1 | Attaching a Cell to the Cell Holder |
|---|---|

Pretreat a measurement sample, charge a cell with the sample, then correctly attach the cell to the cell holder by using the following procedure.

1) Degas the sample and the buffer solution by using an aspirator for 15 minutes or more, and condition them. If degassing is insufficient, bubbles may be accumulated inside a cell during temperature control, and correct measurement may not be available.
2) Charge a cell with the sample by using a pipette.
   * The minimum required sample amount varies on each cell. Check the specifications of the cell to be used.
3) Set a dedicated cell cover.
   In this case, it is ideal when there is no bubble layer between the bottom face of the cell cover and the top face of the sample.
   If there are too many bubble layers, the sample concentration can easily change during temperature control due to evaporation of moisture, etc.
4) Tighten a knurled screw for fixing the cell cover offered as an accessory, and securely fix the cell cover. (Refer to Fig.5.1)
5) If the optical axis of the cell is low (such as in the 115B-QS-10 whose capacity is 400 $\mu$L manufactured by HELLMA), use a bottom-up spacer offered as an accessory. Insert the spacer into a flat spring offered as an accessory, attach the cell on the plate spring, then attach them to the cell holder. (Refer to Fig.5.1)
6) When controlling the temperature to a value lower than the room temperature (for measurement at 10℃ or less), confirm that the connection for $N_2$ purge is all right, then flow $N_2$ gas.

| CAUTION | During measurement, do not remove cell by loosing knurled screw and not draw out cell from cell holder by loosing lever.<br>There is possibility to get out of lid and rush out the sample among cell.<br>Check that the cell has been cooled down till near room temperature before mounting or removing the cell. |
|---|---|

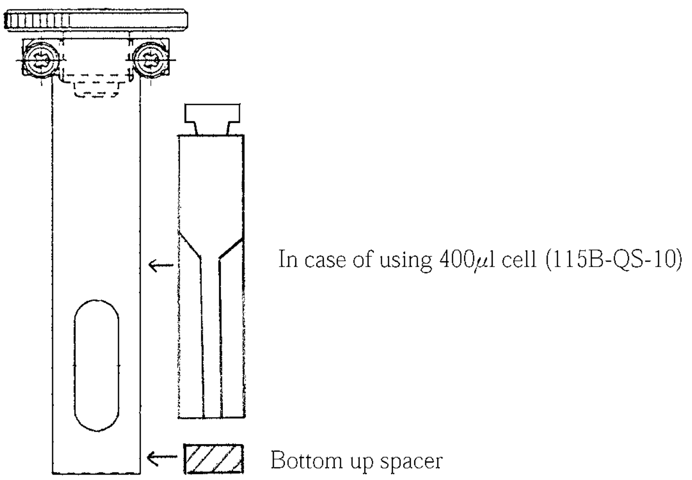



Fig.5.1 Fix cell with Knurled screw

## 5.2 Storing and Cleaning a Cell

(1) Storing a cell

After using the cell and cell lid for measurement, clean them with neutral detergent for optical parts, alcohol or water sufficiently.

After cleaning the cell, fill the cell with distilled water or alcohol and fix the lid then store it.

(2) Cleaning a cell

Use distilled water, alcohol or neutral detergent for optical parts for cell cleaning.

Dirt may adhere to the inner wall in some cases. In such a case, prepare sufficient amount of neutral detergent or following detergent in a beaker or other vessel, put the cell in it and apply it to ultrasonic cleaner.

The specifications of the cleaning liquid and the ultrasonic cleaner are as follows:

(Detergent)

| | | |
|---|---|---|
| Detergent name | : | HELLMA NEX II |
| Manufacturer | : | HELLMA GmbH&Co. |

Shimadzu doesn't deal with this detergent. When purchasing it, contact HELLMA sale department refer to HP ( http://www.hellma.ch/).

(Ultrasound cleaner)

| | | |
|---|---|---|
| Output wavelength | : | 40 KHz |
| Output | : | 100W(Max) |

CAUTION

Use distilled water, alchol, neutral detergent or above special detergent as cleaning detergent.

It is impossible to use fluorine acid, phosphoric acid, sodium hydroxide and potassium hydroxide.

NOTE Handling of "HELLMA NEX II"

Thoroughly read the following cleaning method, then correctly use the cleaning liquid.

Cell cleaning method

(1) Prepare a beaker in which a cell can be sufficiently submerged.

(2) Dilute the cleaning liquid with water (distilled water) until the concentration becomes 0.5 to 2.0%. Make sure to use a container made of glass to pour the diluted cleaning liquid.

(3) Prepare such amount of diluted cleaning liquid that the entire cell is submerged in it.

(4) Submerge the cell in the cleaning liquid, and leave it for the following period of time:

| Cleaning liquid temperature | Period of time |
|---|---|
| 20～25℃ | 120～180 min |
| 30～35℃ | 30～40 min |

(5) After leaving the cell for the period of time shown above, sufficiently wash the inside and the outside of the cell with distilled water.

(6) If the cell is still dirty, clean it (which is submerged in the cleaning liquid) with the ultrasonic cleaner for approximately 20 minutes.

Other Cautions

NOTE 1) The raw cleaning liquid is strong alkali. Take care so that it is not adhered directly on your bear hands. When handling it, use a glass pipette or wear gloves made of thick vinyl.

NOTE 2) When diluting the cleaning liquid, use water (distilled water). Do not use any other liquid.

## 5.3 Installation of Mask

In case of using micro cell type (Cat no. 115B-QS-10, Minimum required sample amount:400 $\mu$ l), it is necessary to install a mask to cell holder (No.15 in table 1.1).

There is a tendency for measurement stability to go down without this mask.

And it is necessary to carry out a baseline correction after installing a mask.

About installation, refer to Fig.5.3 and 5.4.

1) Remove the sample compartment from spectrophotometer.

2) Pull out the cell from cell holder.

3) Remove the cover by loosing knurled-head screws.(refer to Fig.5.3)
4) Insert the mask refer to Fig.5.4.
5) Install the cover again by knurled-head screws.
6) Carry out the baseline correction after setting the blanck cell.

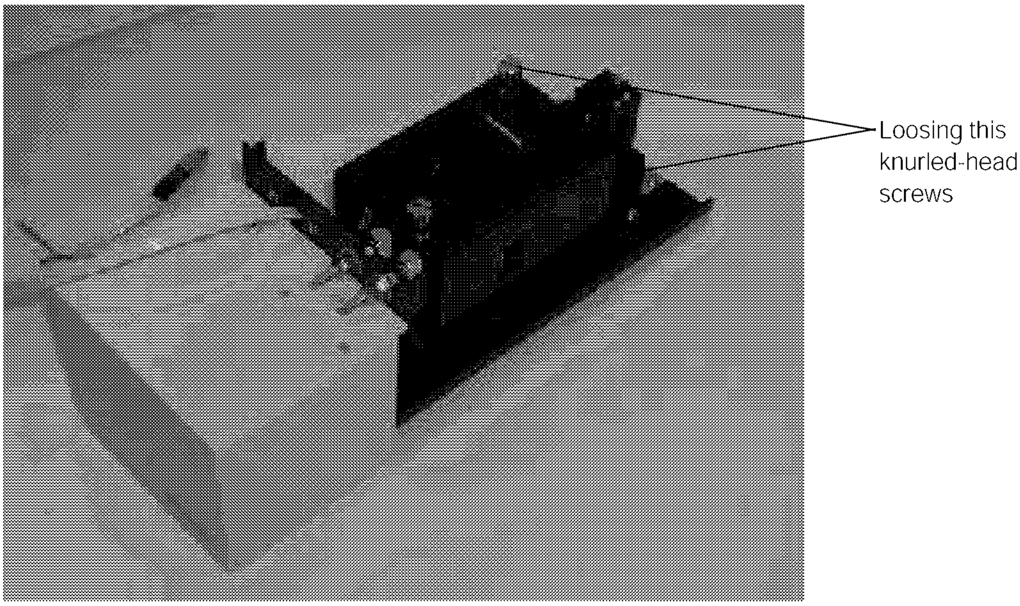



Fig.5.3 Remove the cell cover

Fig.5.4 Installation the mask

# Section 6 How to Use Optional PC Software (Temperature Control in PC)

When the PC software (Tm Analysis Software) is used as an optional function, the temperature control operated by the temperature controller can be executed through the PC.

Contents

| 6.1 | Required Parts |
|---|---|

The optional PC software “Tm Analysis Software” is not available in the MultiSpec-1500 among applicable models shown in “3. Specifications”.

The following parts are required to control this instrument by the PC.

| Parts name | P/N | Quantity |
|---|---|---|
| Tm Analysis Software | 206-57476-91 | 1 |
| RS232C Cable | 200-86408 | 2* |
| USB Cable | 088-52848-32 | 1* |

* When using this optional software, an RS-232C cable connecting the PC and the temperature controller and an RS-232C cable connecting the PC and the spectrophotometer (two cables in all) are required.
If your spectrophotometer is UV-1800 or for the PC type, purchase only one RS-232C cable.
* When using UV-1800, an USB cable connecting the PC and the spectrophotometer (one cable) is required.

In addition, a PC to install the “Tm Analysis Software” is required. Prepare a PC having the following specifications.

| Items | Specifications |
|---|---|
| OS | **Windows XP Professional SP2/Windows Vista Business** |
| Port | 2 COM ports or more * (except for UV-1800)<br>1 COM ports or more and 1 USB ports or more required (UV-1800) |

* If only one COM port is provided in the PC, an extended board (serial parallel card) is required additionally.

| 6.2 | Installing the Software |
|---|---|

1) Insert the CD-ROM of the “Tm Analysis Software” into the CD-ROM drive of the PC.
2) **The procedure to set up the software will start automatically. If the set-up program does not run automatically, double-click the AutoRun.exe in the CD-ROM window.**
3) Please read the information in the welcome box and click Next.
4) In the Choose Destination Location dialog box, use Change key to choose a folder for Tm Analysis, or accept the default folder (Program Files¥Shimadzu¥Tm Analysis). Click Next.
5) In the Ready to Install the Program dialog box, click Install to start install, or use Back key to review or change any of your installation settings.
6) Now, installation is completed.

| 6.3 | Let's Measure |
|---|---|

When the installation is completed, connect the PC and the temperature controller as well as the PC and the spectrophotometer with two RS-232C cables. (In case of UV-1800, one RS-232C cable and one USB cable is required.)
Then, let's measure in accordance with the procedures below.

### 6.3.1 Registering the instruments

First, register the spectrophotometer and the temperature controller to be controlled by the software.

1) Select "Instrument" - "Instrument settings" from the main window.

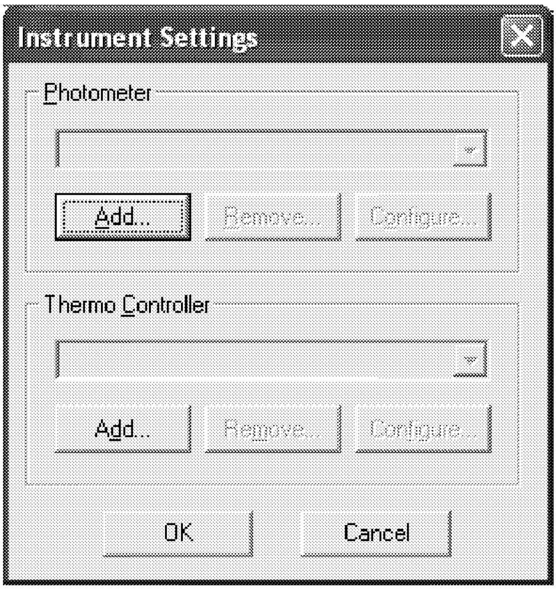



Fig.6.1 Instrument Setting Screen

2) Click [Add] inside the spectrophotometer frame to open the Wizard for adding instrument. Click [Next] in accordance with the instruction on the screen to open the screen to select the spectrophotometer type. Select the series name to be used in measurement, then click [Next]. Enter a proper name to the selected instrument, then finish the registration.

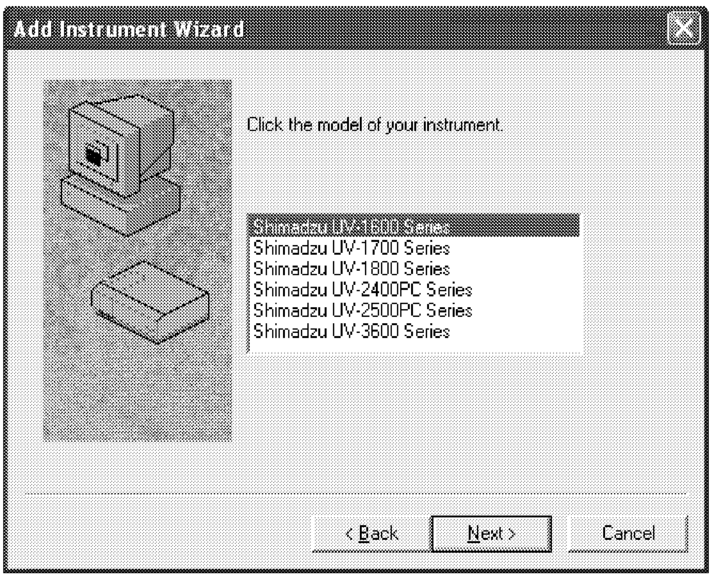





Fig.6.2 Spectrophotometer Selection Screen

3) Click [Add] inside the temperature controller frame. When a screen similar to that for spectrophotometer appears, select the temperature controller and enter a proper name, then finish the registration.

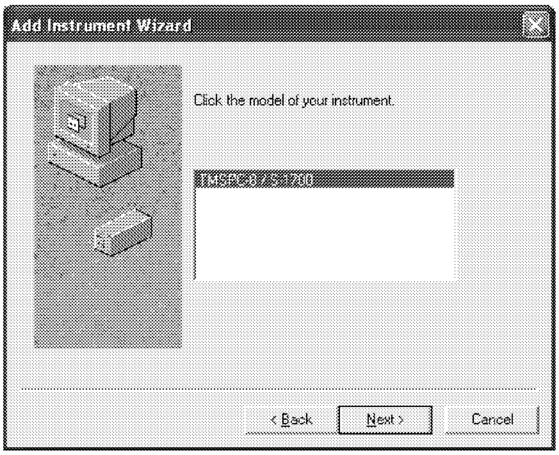



Fig.6.3 Selecting the Temperature Controller

4) At the end, click [Settings], and set the COM ports of the selected spectrophotometer and temperature controller. Check the status of each connection port in the PC, then select ports.

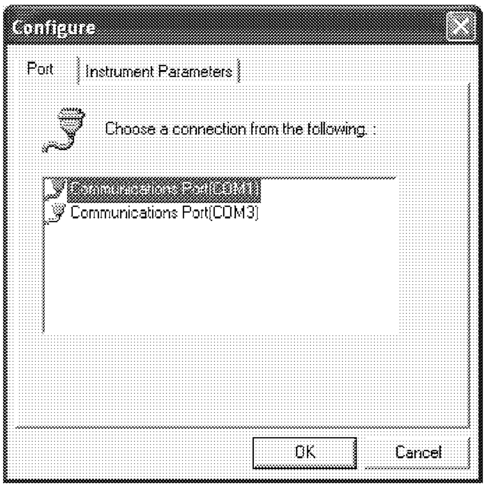



Fig.6.4 Setting the Ports

### 6.3.2 Let's connect the spectrophotometer

1) When the settings so far are finished, turn on the power of the spectrophotometer and the temperature controller.
2) Click the "Connect" icon at the bottom of the window to start initialization in accordance with the set spectrophotometer.

   If the UV-1600/1700/1800 is used as the standalone type, the following procedure is all right:
   Initialize only the spectrophotometer.
   Select the "External Control" mode.
   Click the "Connect" icon.
3) When the initialization is normally completed, click [OK].
4) Now, preparation for temperature control is completed.

### 6.3.3 Let's create temperature programs

Let's create temperature programs, and start measurement actually.

When this software is used, the spectrophotometer measures the absorbance at each "Measurement Interval" in the program.

Let's create programs.

Click this icon

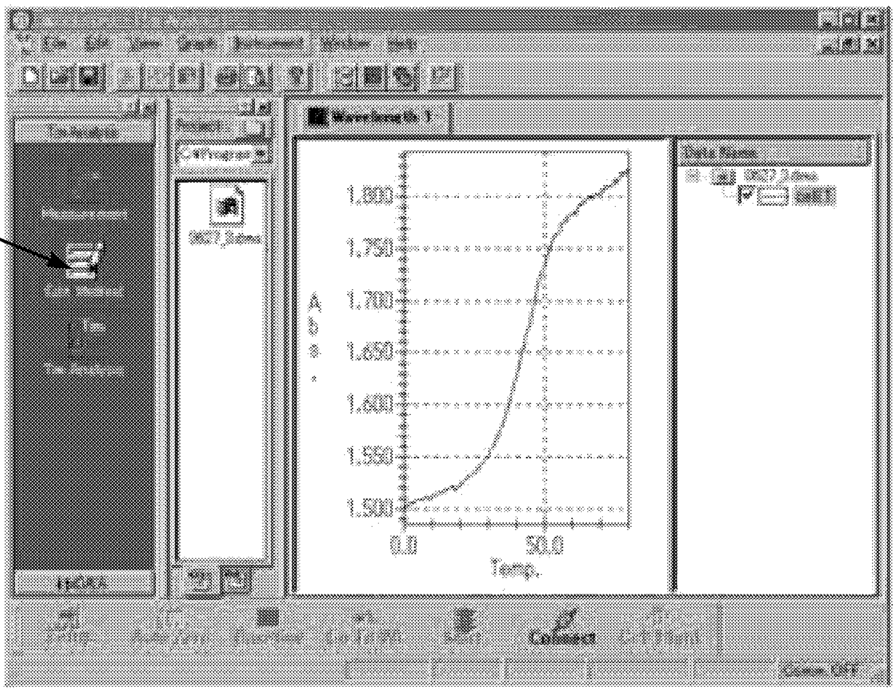



Fig.6.5 Tm software main screen

Click the "Edit Method" icon in the left row of the screen to display the following window to create temperature programs.

Click here

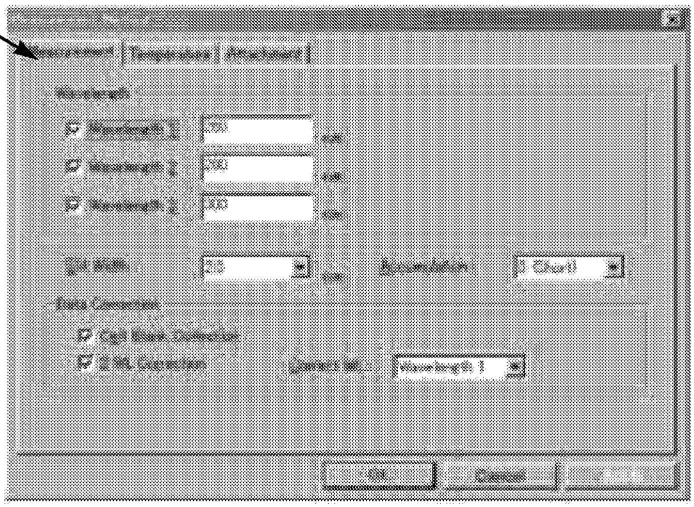


Fig.6.6 Measurement Method Setting Screen

On the "Measurement" worksheet, set parameters concerning absorbance measurement in the spectrophotometer.

"Meas. Wavelength" ——— Up to three types of measurement wavelengths can be specified. Check the check box of "Wavelength 1/2/3", then enter a wavelength.

"Slit Width" ——— Select the slit width used in measurement. In the UV-1600/1700/1800, the slit width is fixed.

"Integ. Time" ——— Select the signal integrating time in measuring the absorbance. As a value is larger, noise is less but the time is longer. Usually, "0" (default value) can be used. However, when measuring a sample whose absorbance is more than 1.5 Abs, enter a large value.

"Cell Blank Correction" ——— Not used.

"Two wavelength correction" ——— This parameter is effective only when two or three wavelengths are used in measurement. When the check box is checked, a measurement file in which the "Correction wavelength" data is subtracted from the "Measurement wavelength" data is automatically created.

"Correction Wavelength" ——— Enter the correction wavelength to be used when "Two wavelength correction" is executed in measurement.

> NOTE
>
> **Integ. Time**
>
> This parameter indicates the time duration in which the spectrophotometer reads data. When a large value is entered, the variation is small in the acquired data. It is usually set to "0".

> NOTE
>
> **Two wavelength correction**
>
> In the same way as cell blank correction, this parameter helps to acquire data of fine precision. Use this parameter to correct the effect of interfering elements, scattering caused by turbidity as well as the effect of generated bubbles on the absorbance. The absorbance is measured at the "Correction wavelength" in the same way as the "Measurement wavelength", then the measurement wavelength data subtracted by the correction wavelength data is used in data processing.

Next, set parameters for temperature control in the "Temperature" worksheet.

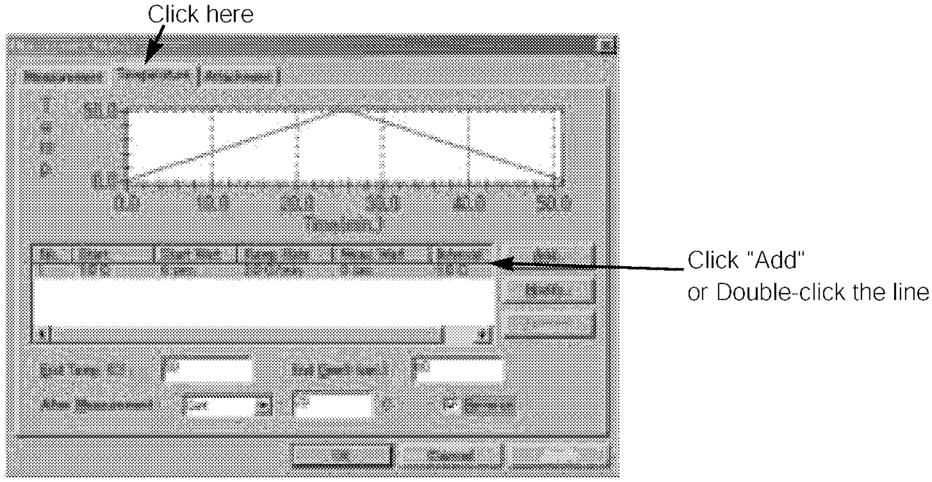



Fig.6.7　Temperature Program Setting Screen

Click the [Add] button to display the "Add temperature schedule" screen.

In this software, one temperature program can be divided into 10 partitions maximum.

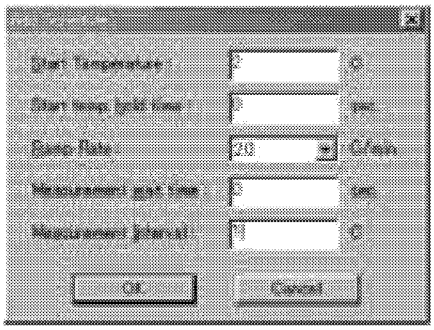


Fig.6.8　Temperature Schedule Setting Screen

When the setting window is open, enter each set value while referring to the explanation below.

| | |
|---|---|
| Start Temp | : Enter the temperature at which measurement is started. The range is from 0 to 110℃, and the unit is 0.1℃. |
| Start Temp hold time | : Enter the time until measurement is started. If the current temperature is considerably different from the start temperature, rather long time should be set while considering the time for temperature movement. Enter a proper value while referring to NOTE below. The range is from 0 to 18,000 sec, and the unit is 1 sec. |
| Ramp Rate | : Enter the temperature increase/decrease speed between the start temperature and the end temperature. The temperature speed can be entered in 12 steps from ±0.1 to 5.0℃/min. The code of the temperature speed is automatically set based on the start temperature and the end temperature. |
| Measurement wait time | : Enter the time in which the temperature is retained at each measurement interval. The range is from 0 to 18,000 sec, and the unit is 1 sec. |
| Measurement Interval | : Enter the temperature interval in measurement. The range is from 0.1 to 20℃, and the unit is 0.1℃. |

| NOTE | Start Temp. Retention Time<br>If the current temperature is considerably different from the start temperature, the retention time should be set long. As reference, if the current temperature is 20℃ and the start temperature is 0℃, the retention time is as follows:<br>[(20℃ - 0℃) / 5℃] × 60 + 600 sec = 840 sec |
|---|---|

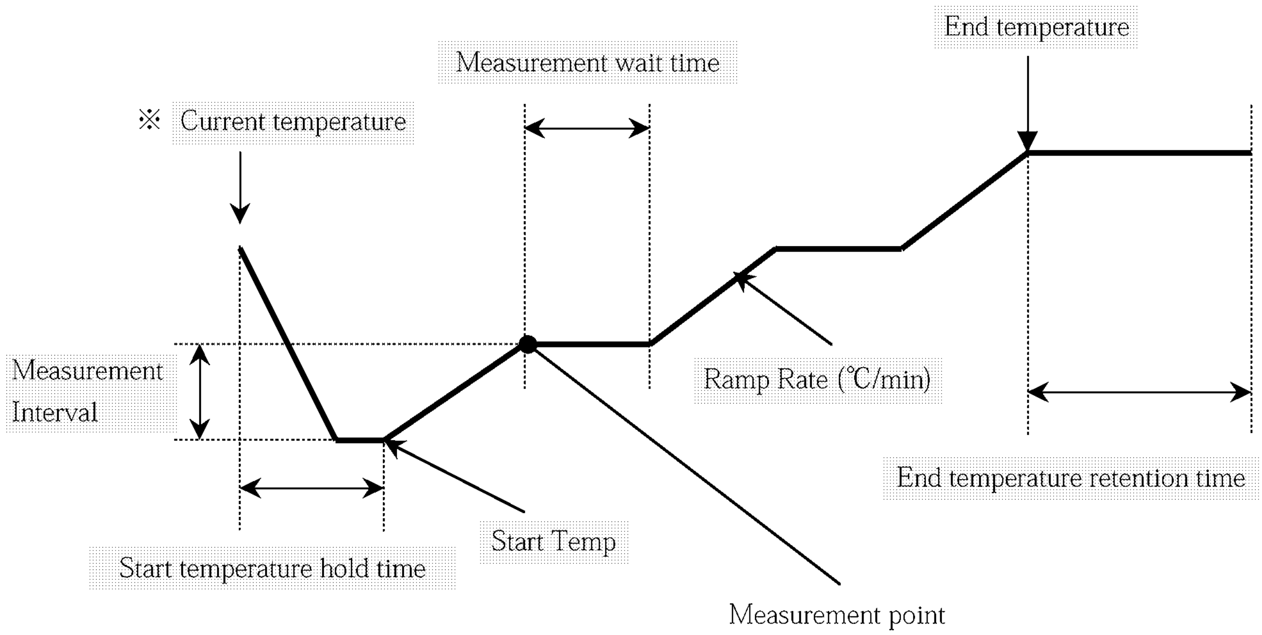



Fig.6.9 Outline of Temperature Program

| NOTE | When making a program in which the temperature speed is changed in each temperature range (for example, 1.0℃/min in the range from 0 to 10℃ and 5.0℃ /min in the range from 10 to 20℃), repeat the "Add" operation and make several temperature programs. |
|---|---|

Other buttons on the screen have the following functions. Refer to the explanation in creating temperature programs.

"Modify" ——Changes an existing program.
To change the program, double-click line to be changed.

"Remove" ——Deletes an existing program.
Click a line to be deleted to select it, then click this button.

"End Temp." ——Enter the temperature at which measurement is finished.

"End Dwell" ——Enter the time in which the end temperature is retained.

"After Measurement" ——Set the temperature control operation to be executed after all temperature programs are finished.
When the temperature is set, enter a temperature value in the box.

"Reverse" ——Turn on the check box to repeat measurement, after reaching the end temperature once, until the start temperature. The temperature speed is controlled at the preset temperature speed with a reverse code.

Next, set the sample compartment type to be used.

Click [Device], and select the cell holder type.

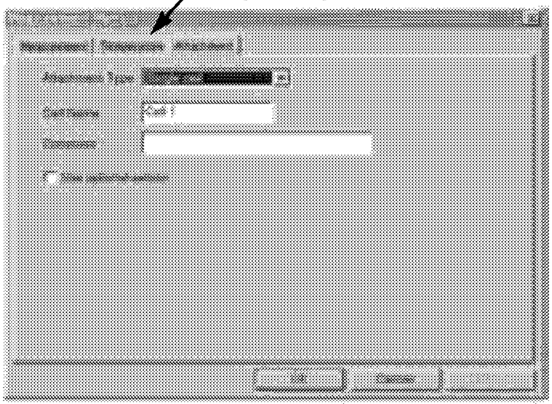


Fig.6.10 Device Setting Screen

Select "Single Cell" in "Attachment Type". Enter characters to "Cell" and "Comment".

* Do not turn on "Use optional sensor".

Now, setting of the temperature programs is completed.

Set a sample to the cell. Set the cell to the holder, then click the [START] button at the bottom of the screen. Then, the preset programs will be started, and the absorbance will be plotted on the graph.

# 6.4 Outline of Optional PC Software

## 6.4.1 Main screen

A series of measurement method is described in the previous section. This section describes other functions.

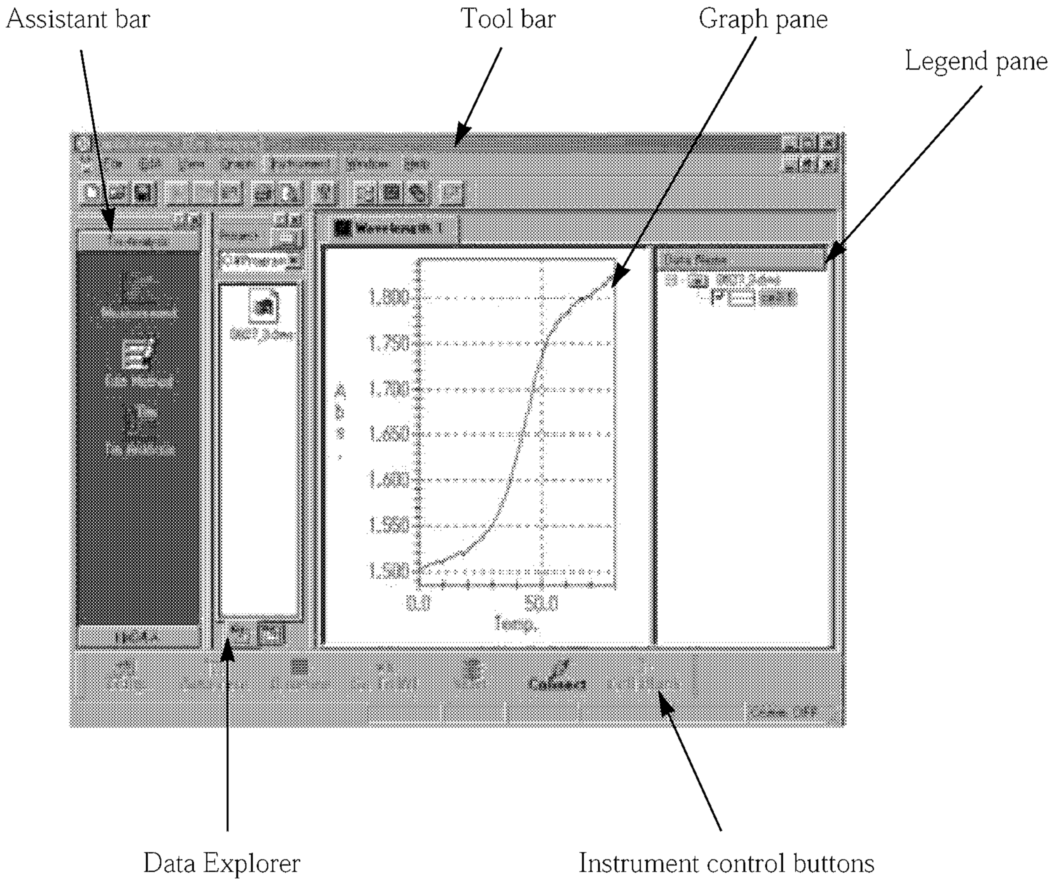



Fig.6.11 PC Software Main Window

"Assistant bar"

Switches the measurement screen and the method edit screen.

* "Tm Analysis" is not used.

"Data Explorer"

Data stored in a preliminary specified folder are displayed with icons. Each data can be easily displayed by dragging and dropping it to the Graph pane.

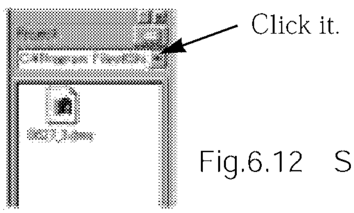



Fig.6.12 Setting the Data Storage Folder

"Instrument control buttons"

Controls the spectrophotometer are mainly displayed with icons.

"Temp."

Click it to display the screen which sets the cell to arbitrary temperature.

Enter the target temperature, the cell temperature increase/decrease speed during movement and the retention status at the target temperature, then click [OK] to start control.

"AutoZero"

Sets the displayed absorbance in the spectrophotometer to 0 (zero). Use this command in the blank status before measurement.

"Baseline"

Sets the baseline in the preset wavelength range. Before measuring a sample, measure the baseline in the blank status or in the status in which the buffer is charged in the cell.

"GoTo WL"

Moves to an arbitrary wavelength.

"START／STOP"

Starts or stops the "temperature vs. absorbance" measurement. When measurement is finished or stopped, the file save screen to save the measurement data appears.

"Connect/Disconnect"

Connects or disconnects the spectrophotometer to/from the PC and the temperature controller. Use this command while the power of the spectrophotometer and the temperature controller is ON.

"Cell Blank"

Available when "Cell Blank Correction" is set to ON on the measurement method setting screen. Use this command before measurement to exclude the effect of the buffer absorbance on a measurement sample. If [Start] is clicked while cell blank has not been measured, a message appears to notify that cell blank has not been measured.

"Graph pane"

When measurement is started, the "temperature vs. absorbance" data is plotted in this area. Double-click the right mouse button on this pane to apply auto scaling.

"Legend pane"

Indicates the display or hide status of the currently active data.

* The "Edit" and "Window" menus on the Menu bar are not used in this instrument.

### 6.4.2 "Menu bar" - "File" menu

Select "File" on the Menu bar to display the pull-down menu below. The following items are for controlling data files, concerning printing, etc.

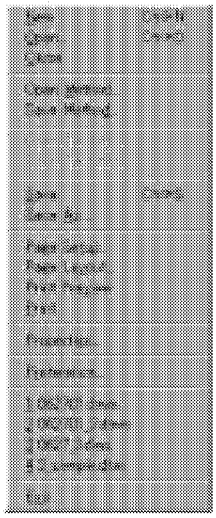


Fig.6.13 "File" menu

"New"

Creates a new Graph pane.

When this item is selected, all data displayed on the Graph pane and the Legend pane are deleted. If some data has not been saved, a message appears to notify that some data has not been saved.

"Open"

Loads a saved data file to the Graph pane and the Legend pane.

In this instrument, files with the extension ".dms" are handled as S-1700 data.

* Files with the extension ".dmm" are data of another device, and not handled in this instrument.

"Close"

Select this item in the "Measurement" or "Tm Analysis" mode to close the Graph pane and the Legend pane.

"Open Method"

Reads a saved measurement method file.

"Save Method"

Saves the current measurement parameters as a measurement method file.

"Open Tm Table"

Not available in this instrument.

"Save Tm Table"

Not available in this instrument.

"Save"

Saves the measurement data with the file name of an existing data on the Graph pane. (The existing data is overwritten.)

"Save as"

Saves the measurement data on the Graph pane with a new name.

"Page Setup"

Sets various parameters of the printer to be used in printing.

"Page Layout"

Displays the screen shown in Fig.6.14 to set the print layout.

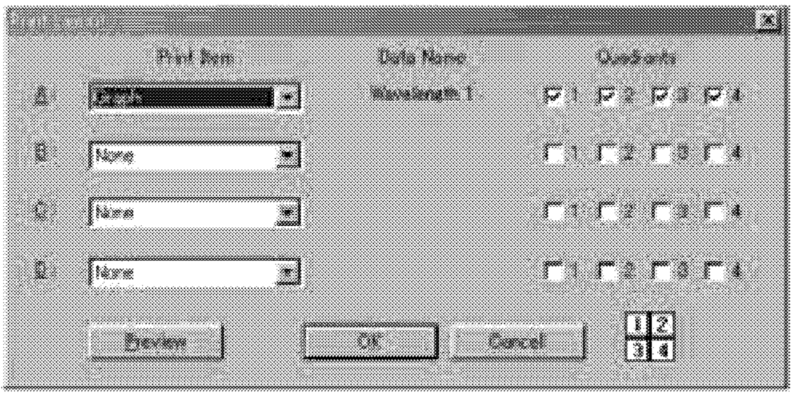



Fig.6.14　Print Setting Screen

Print Items : Select items to be printed in the selected partition.

Select either one among "No", "Graph", "Parameters" and "Text".

* Tm table is not set in this instrument.

(Graph): Displays the screen to select the measurement wavelength of the data to be printed.

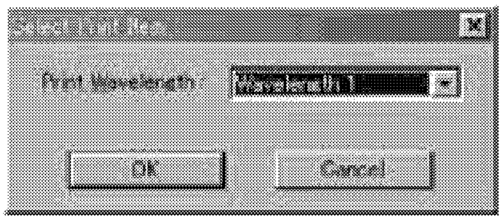



Fig.6.15　Screen when "Graph" is selected

(Tm Table): Not set in this instrument.

(Parameters): Select measurement parameters to be printed.

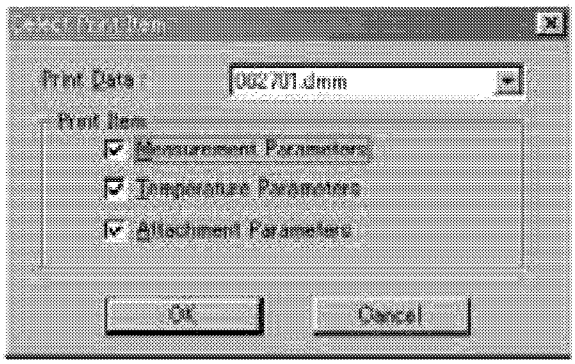



Fig.6.16　Screen when "Parameters" is selected

(Text): Displays the contents of the edited text on the clipboard.

Quadrants : Set the destination of layout of the selected print items.

If partitions are not selected, if three partitions are set for one print item, or if partitions are set in diagonal locations, an error message is displayed.

Preview : Displays the preview to check the set print image.

* Print settings can be set for only one page in this software.

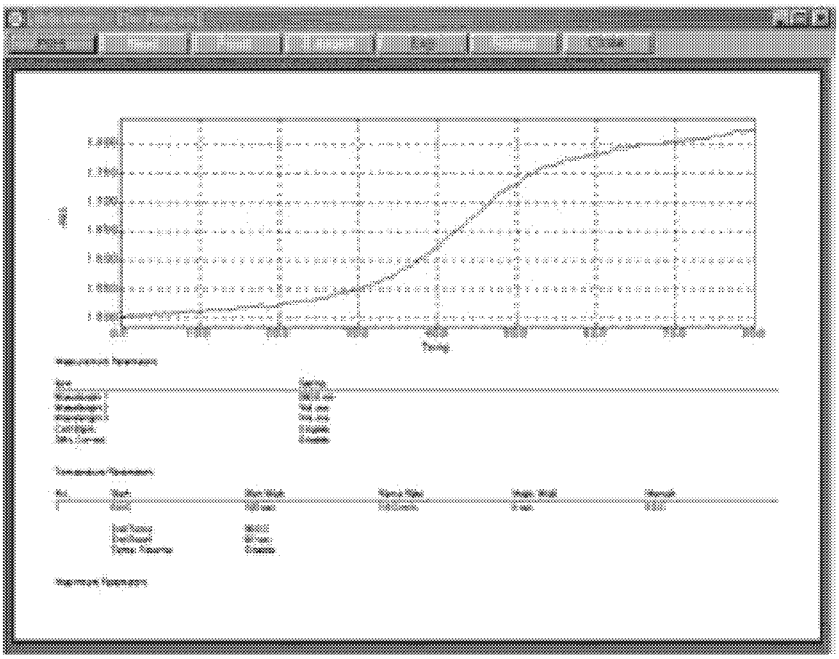


Fig.6.17　Preview Screen

"Print Preview"

Confirm the print image set in "Print Layout Settings". This item has the same function as "Preview" in "Print Layout Settings", but can be directly selected from the "File" menu.

"Print"

Prints the image set in "Print Layout Settings".

"Properties"

Confirm the attributes of the data currently displayed on the Graph pane. The following contents can be confirmed.

1) Spectrophotometer measurement parameters

2) Temperature control program

3) Use setting in cell position

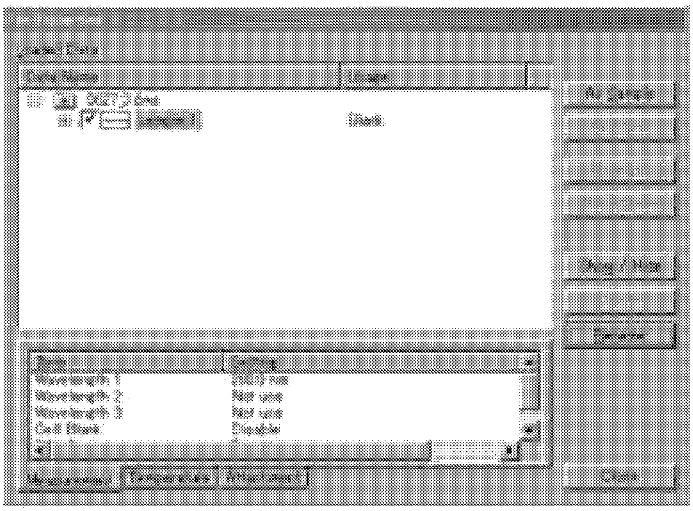


Fig.6.18　Properties Selection Screen

“Preference”

Set the attributes of “View”, “Initial Line” and “Text”.

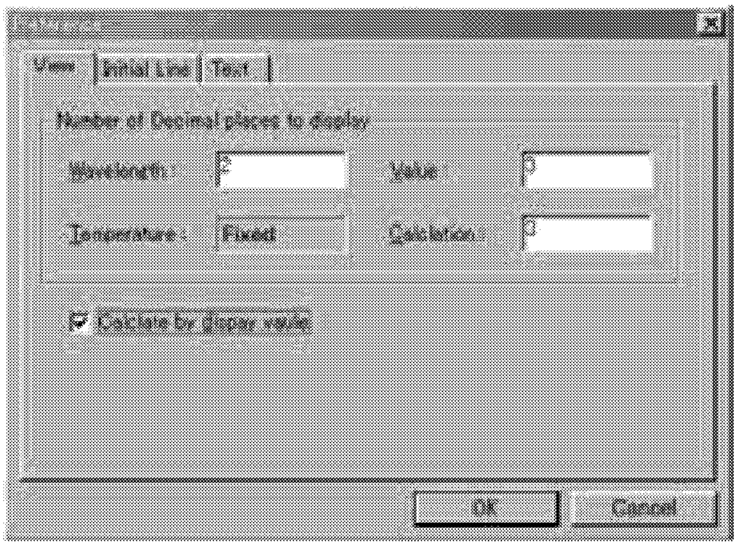


Fig.6.19 Configuration Setting Screen

View : Set the number of digits after the decimal point of the wavelength or the absorbance displayed on the status bar as well as calculated values displayed in the “Tm Analysis” mode. When data is saved in the text format (extension: .txt), values are saved in accordance with the number of digits set here.

Initial Line : Set data displayed on the Graph pane.

Text : Set the character string used to delimit the data field when data is saved in the text format. The setting is effective in post processing in the Excel, etc.

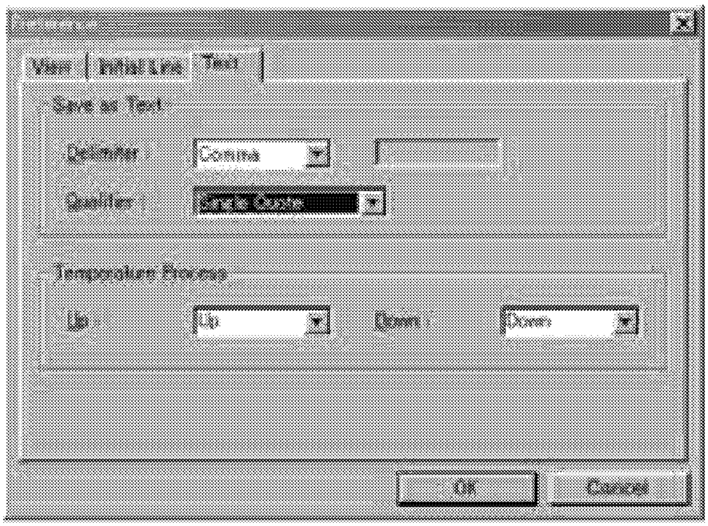


Fig.6.20 Text Configuration Setting Screen

On the pull-down menu, select the delimiting character and the character string quotation added when text is saved.

The names set in “Go” and “Come” inside the temperature stroke frame are effective when “Temp. Reverse” is set to ON in the temperature program created on the measurement method setting screen. The name set on the pull-down menu is handled as the default, and name is added to a file in each temperature process.

### 6.4.3 "Menu bar" - "View" menu

Select "View" on the Menu bar to display the pull-down menu below. On the "View" menu, switch the Graph pane, and set the display or hide status of each function tool on the screen.

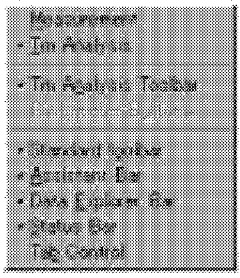


Fig.6.21 "View" Menu

"Measurement"

Select the mode to measure the absorbance.

"Tm Analysis"

Not used in this instrument.

"Tm Analysis Tool bar"

Not used in this instrument.

"Photometer Buttons"

Use these command buttons to mainly manipulate the spectrophotometer.

The "Temp." command jumps the cell temperature to an arbitrary set value, then holds it (or does not hold it depending on the setting).

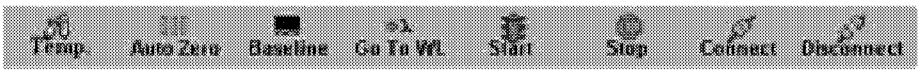



Fig.6.22 Instrument Control Buttons

"Standard toolbar"

Switches the display or hide status of icons of frequently used functions such as "Save" and "Print". Click an icon to execute a corresponding command.

Fig.6.23 Standard tool bar

"Assistant Bar"

Switches the measurement mode and the method edit mode by clicking an icon corresponding to a desired mode.

"Data Explorer Bar"

If a folder saving data has been specified in advance, all data saved in the folder are displayed in the shape of icons. Drag a displayed icon and drop it to the Graph pane to display the corresponding data.

"Status Bar"

This bar indicates the current status of the temperature controller and the spectrophotometer. The indicated contents automatically change in accordance with the manipulation.

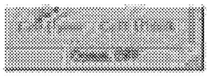


Fig.6.24 Status Bar

"Tab Control"

Not used in this instrument.

### 6.4.4 "Menu bar" - "Graph" menu

Select "Graph" on the Menu bar to display the pull-down menu below. Set the graph displayed on the Graph pane.


Fig.6.25 "Graph" menu

"Auto Scale"

This item applies auto scaling to the data displayed on the Graph pane in accordance with the auto scale size set in "Settings" - "Line Type" shown below.

"Settings"

"Line" ——Set the data mark, the line type, the line width and the line color.

"Limits" ——Set the display range of the X axis (absorbance) and the Y axis (temperature). In addition, the magnification for "Auto Scale" can be set also.

"Appearance" ——Set the background color of the Graph pane.

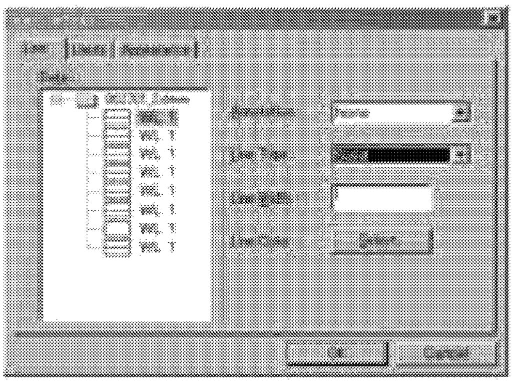


Fig.6.26 Graph Setting Screen

### 6.4.5 "Menu bar" - "Instrument" menu

Select "Instrument" on the Menu bar to display the pull-down menu below. Parameters concerning measurement can be set, and control commands for the spectrophotometer can be executed.

Same commands are offered also as Instrument control buttons.

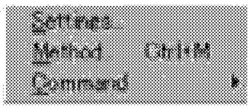



Fig.6.27 "Instrument" Menu

"Settings"

Sets the connection condition of the spectrophotometer and the temperature controller.

For details, refer to "6.3.1 Registering the instruments".

"Method"

Creates temperature programs and perform various settings for measuring the absorbance.

For details, refer to "6.3.3 Let's create temperature programs".

"Command"

Mainly manipulate the spectrophotometer. These commands can execute the same functions as those offered by the instrument control buttons displayed at the bottom of the screen.

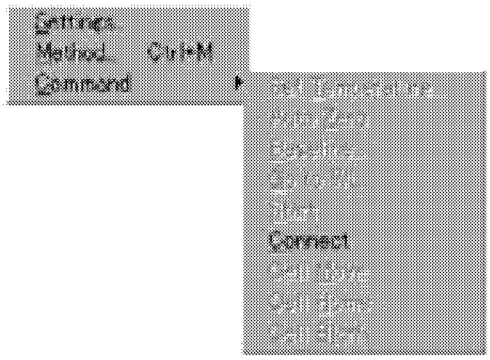



Fig.6.28 "Command" Menu Items

"Set Temperature" —— Opens the temperature screen.
When entering a temperature value, the control toward the temperature is started.

"Auto Zero" —— Adjusts the zero point of the spectrophotometer.

"Baseline" —— Corrects baseline.
Enter the start wavelength and the end wavelength.

"Go to WL" —— Opens the screen to enter the wavelength.
When a wavelength value is entered, the setting toward the wavelength is executed.

"Start" —— Executes a temperature program in accordance with the measurement parameters set in the method edit mode.

"Connect / Disconnect" —— Disconnects the connection to the spectrophotometer.

"Cell Move" —— Moves the cell by one position at a time.

"Cell Home" —— Moves the cell to the position 1.

"Cell Blank" —— Acquires the blank correction data before measuring a sample.

| 6.5 | Error Message List |
|---|---|

When a problem has occurred while a temperature program is created in the "Edit Method" mode or while measurement is executed, an error message shown below is displayed. Refer to the countermeasures, and enter a proper set value.

Error messages displayed while a temperature program is created

| Message | Cause | Countermeasures |
|---|---|---|
| The setting of measurement end temperature is not correct. | ① The measurement start temperature is equivalent to the measurement end temperature.<br>② The difference between the measurement start temperature and the measurement end temperature is smaller than the measurement interval. | ① These temperatures cannot be equivalent.<br>② Set the measurement interval less than the difference between the measurement start temperature and the measurement end temperature. |
| The start temperature of temperature schedule No.2. | ① The measurement start temperature is equivalent between the No. 1 and the No. 2.<br>② The difference in the measurement start temperature between the No. 1 and the No. 2 is smaller than the measurement interval in the No. 1. | ① A same start temperature cannot be set in consecutive schedule lines.<br>② In the No. 1, set the measurement interval less than the difference in the measurement start temperature between the No. 1 and the No. 2. |
| If the temperature schedule of increase/decrease is mixed, the Tm value cannot be calculated. | This error message is for any device other than this instrument. | Start the measurement. |
| Sample cell is not set. | 8 Micro Multi Cell" is selected on the device setting screen. | Select "Single Cell". |

Error messages displayed while measurement is executed

| Message | Cause | Countermeasures |
|---|---|---|
| Measurement of blank cell has not completed. | Click [Start] before Cell Blank Correction is executed. | If "Cell Blank Correction" is set to ON, execute it before clicking [Start]. |
| Cooling water does not flow. | Cooling water is not flowing. | Cooling water should be flown to protect Peltier elements. The block temperature is abnormal.Turn off the power once, then confirm that the reflux water is properly connected. |
| Over heat error. | Overcurrent has flown in Peltier elements. | Turn off the power once, then connect the instrument again.If the same error occurs again, the main body has failed. Contact our service personnel. |
| Not found temperature sensor. | The signal cable on the side of the sample compartment is disconnected. | Turn off the power of the PC and the temperature controller once. Confirm that the signal cable is properly connected, turn on the power again, then execute communication. |

# Section 7 Maintenance and Inspection

Contents

| 7.1 | Troubleshooting |
|---|---|

If a phenomenon shown in the table below has occurred during measurement and correct operation is not available, perform inspection in accordance with the explanation.

| Phenomenon | Inspection points | Countermeasures |
|---|---|---|
| Even when the power is turned on, the LEDs in the controller do not light. | Is the power cable connected correctly?Are the power fuses not blown out? | If fuses are blown out, replace them while referring to 7.2. |
| A measured value in the spectrophotometer is abnormal. | Is sample full more than min value? Are bubbles not accumulated inside the cell?<br>Charge the cell with pure water, then acquire the "cell holder base line" or the "base line" again. | Clean the cell.<br>Securely degas the sample.<br>If the spectrometer does not operate normally even after the inspection, contact our service personnel. |
| The evaporation amount in the cell is large. | Is the cover pushed in securely?<br>Is the cell hole free from damages? | Securely push in the cover, then start measurement.<br>If the cell is damaged, replace it. |
| The cell cannot be clean. | Perform ultrasonic cleaning with the cell submerged in sufficient amount of neutral detergent or alcohol. | Securely clean the cell after measurement.<br>If the cell will not be used for measurement, securely dry it or fill it with pure water or alcohol, then store it. |
| The sheet key is disabled. | Is the power turned on correctly? | If the temperature controller does not operate normally even after the inspection, contact our service personnel. |
| A temperature value is wrong. | If an external sensor is installed, is it correctly inserted into the cell?Are the cables correctly connected? | If a displayed value is clearly abnormal even after the inspection, contact our service personnel. |
| The temperature does not decrease sufficiently. | Is the cooling water flown?<br>Is the cooling water set to the correct temperature (20±2℃) ? | If the temperature controller does not operate normally even after the inspection, contact our service personnel. |

| 7.2 | Replacing Fuses |
|---|---|

| | |
|---|---|
| WARNING | Caution on electric shock<br>You may get electrical shock. Before replacing a fuse, turn off the power, then disconnect the power cable. |

If the LEDs on the controller do not light even when the power is turned on, fuses in the power switch may be blown out. In such a case, replace blown fuses with specified ones offered as accessories by using the following procedure.

(1) Turn off the power.
(2) Disconnect the power cable from the power connector.
(3) Pull out the fuse holder by inserting the screwdriver as shown in the figure below.

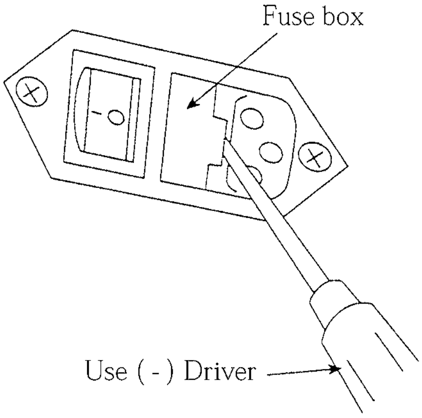



Fig.7.1 How to Open the Fuse Holder

(4) Remove a blown fuse from the fuse holder. Insert a new fuse instead. Replace only a blown one. Insert a new fuse into the holder.
(5) Insert the fuse holder until a clicking sound is heard.

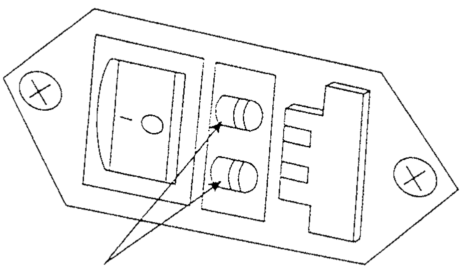



Fig.7.2 Inside of Fuse Holder

NOTE After replacing fuses, turn on the power switch, then confirm that the temperature controller operates normally.

CAUTION Fuse rating

When replacing a fuse, make sure to replace it with one of the designated specifications.

Power voltage for 100～120V (P/N 072-02004-19)

250V 2A T type (5×20)

for 200～240V (P/N 072-02004-17)

250V 1.25A T type (5×20)

# 7.3 Replacing Circulating Water Tubes

Replace the water circulation (for peltier protection) tube every three years. Tube damage can cause an accident of main unit. At first, remove the tube clump and replace to new tubes.

Note that replacement is done after turning off the main power switch and pulling out the power cable.

Silicon tube P/N 016-31350-74

Necessary Length : 210±10mm (one piece)

280±10mm (one piece)

80±10mm (one piece)

Hose Band P/N 072-60320-01 (sixth pieces)

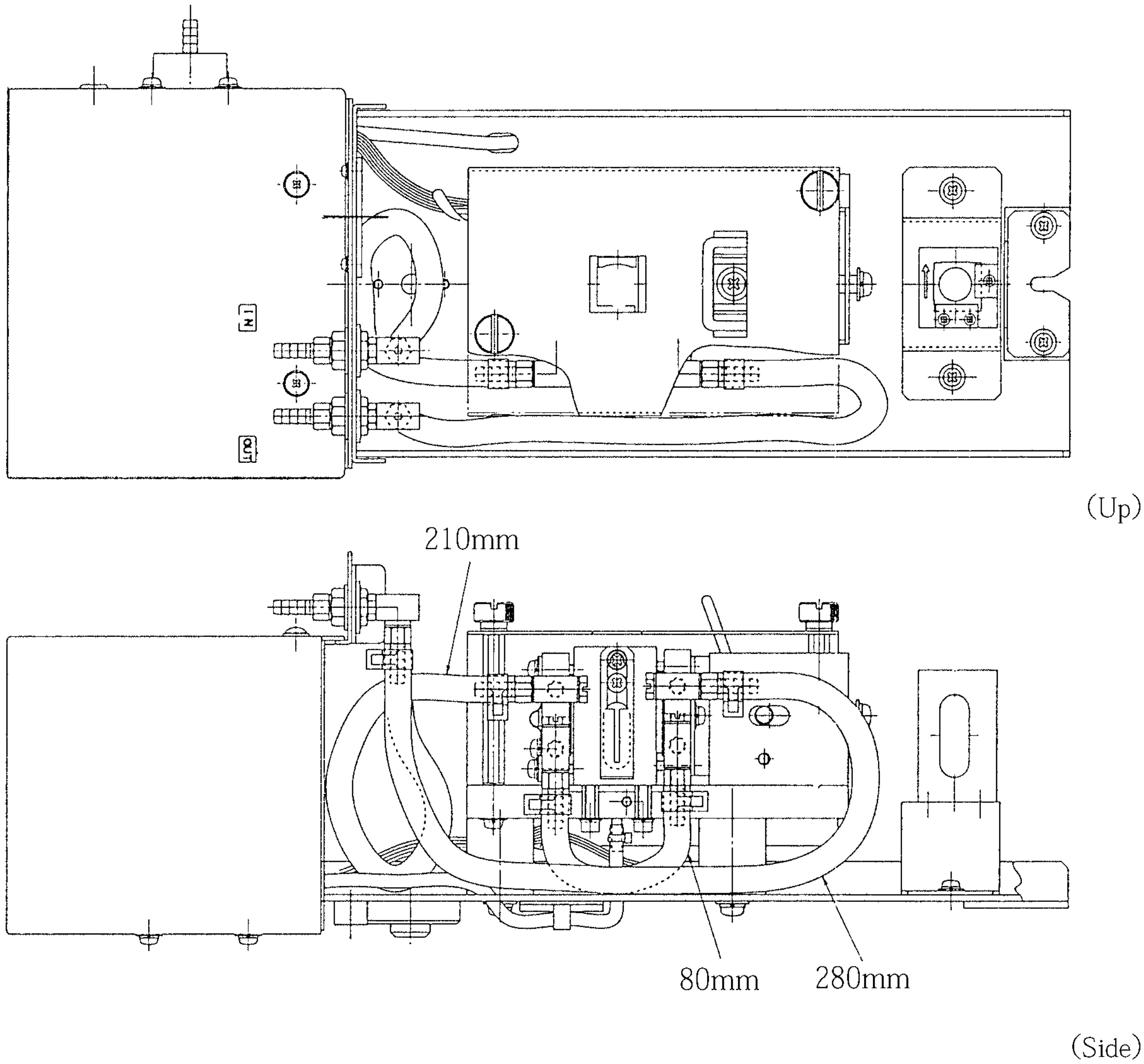



Fig.7.3 Replace the circulation tube

## 7.4 Consumables and Maintenance Parts List

| Name | P/N |
|---|---|
| Bottom-up spacer | 206-56225 |
| Cell Cover Assy | 206-56224-91 |
| Stirrer tip | 046-00619-01 |
| Stirrer controller unit | 208-97250-21 (100～120V)<br>208-97250-24 (220～240V) |
| Stirrer | 208-97250-11 |
| Fuse (for 100～120V) 2A | 072-02004-19 |
| Fuse (for 220～240V) 1.25A | 072-02004-17 |
| AC power cable (for 100～120V) | 071-60814-01 |
| AC power cable (for 220～240V) | 071-60814-05 |

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